



Hippocampus and prefrontal cortex differentially shape memory responses in visual cortex



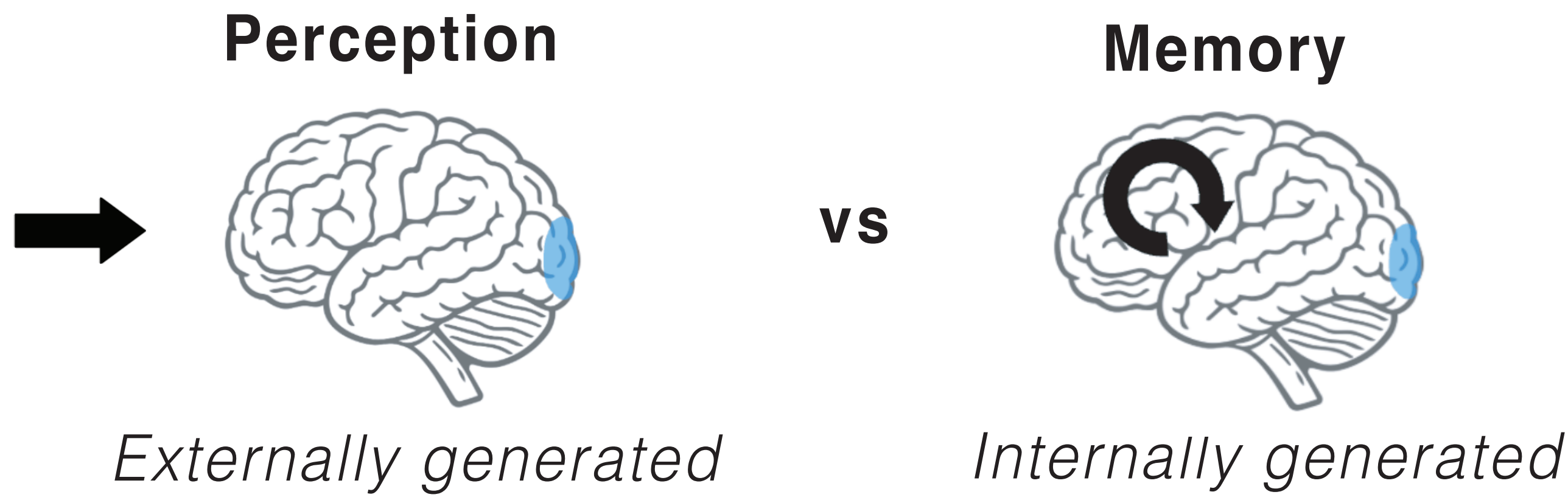
Aditi Narayan, Robert Woodry, Clayton E. Curtis, Jonathan Winawer

Department of Psychology, New York University; Center for Neural Science, New York University

How is neural activity during memory generated in sensory cortex?

Sensory cortex is activated when we see things (externally generated; **perception**), when we briefly hold things in mind (internally generated; **working memory**), and when we recall things from long ago (internally generated; **long-term memory**).

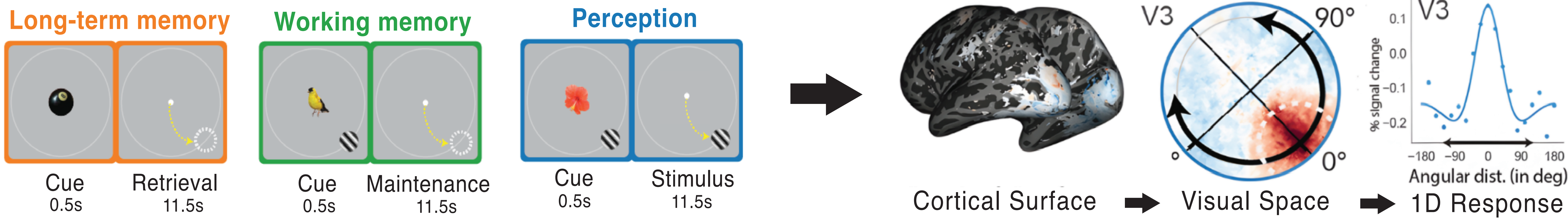
Here, we asked what brain regions are involved in **long-term memory**, thought to rely on hippocampus, and **working memory**, thought to rely on prefrontal cortex.



We measured visual cortex activity during perception, working memory, and long-term memory

Subjects viewed, maintained in working memory, or retrieved from long-term memory the peripheral target associated with 16 unique objects, and then made a saccade to the target location.

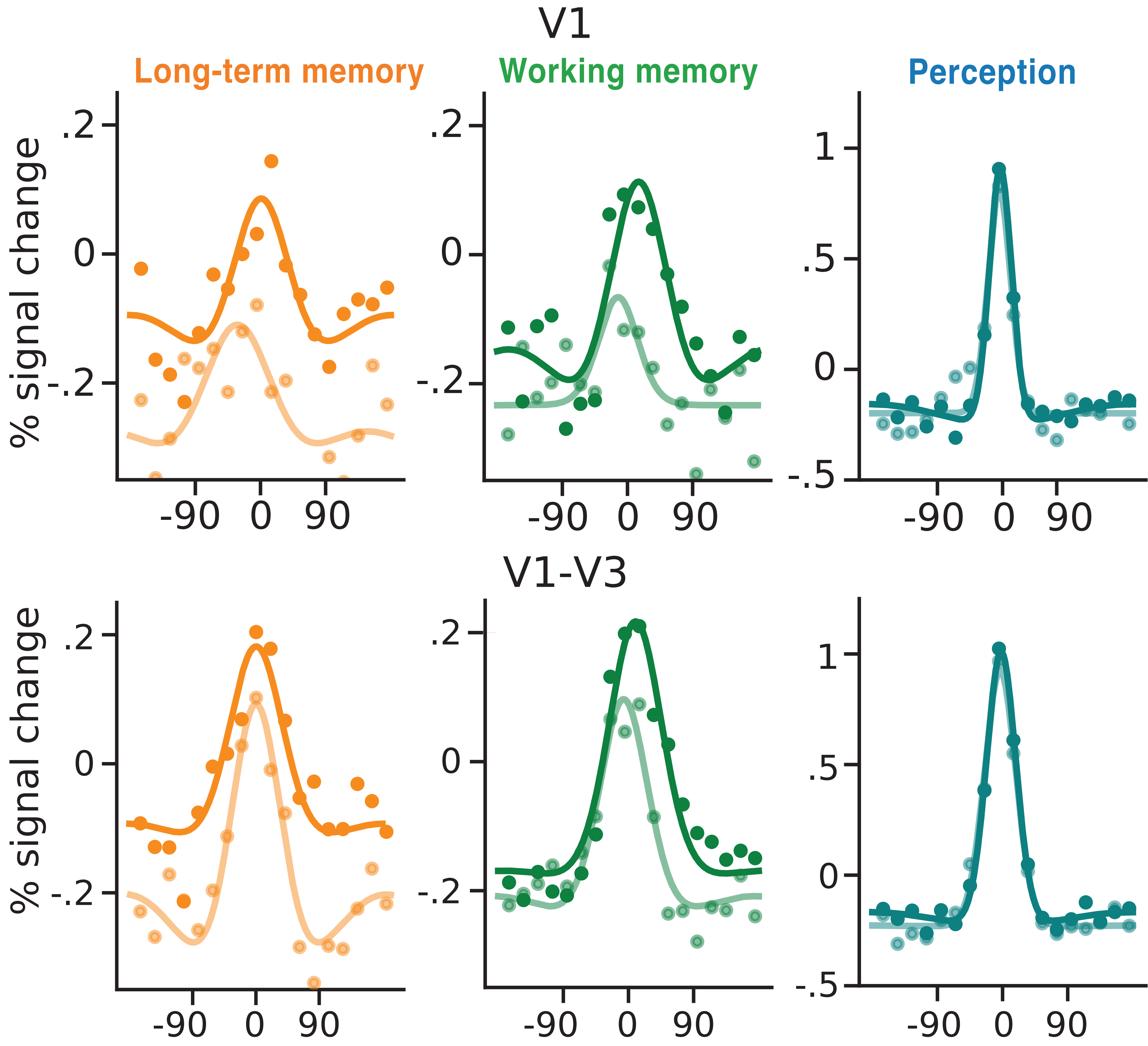
Visual cortex contains an actual map of the retina, adjacent points in images project to nearby location in cortex. This map enables us to project brain activity back into visual space.



We separately quantify visual cortex responses as binned by activity in these higher order regions and conduct a trial-by-trial analysis of these responses

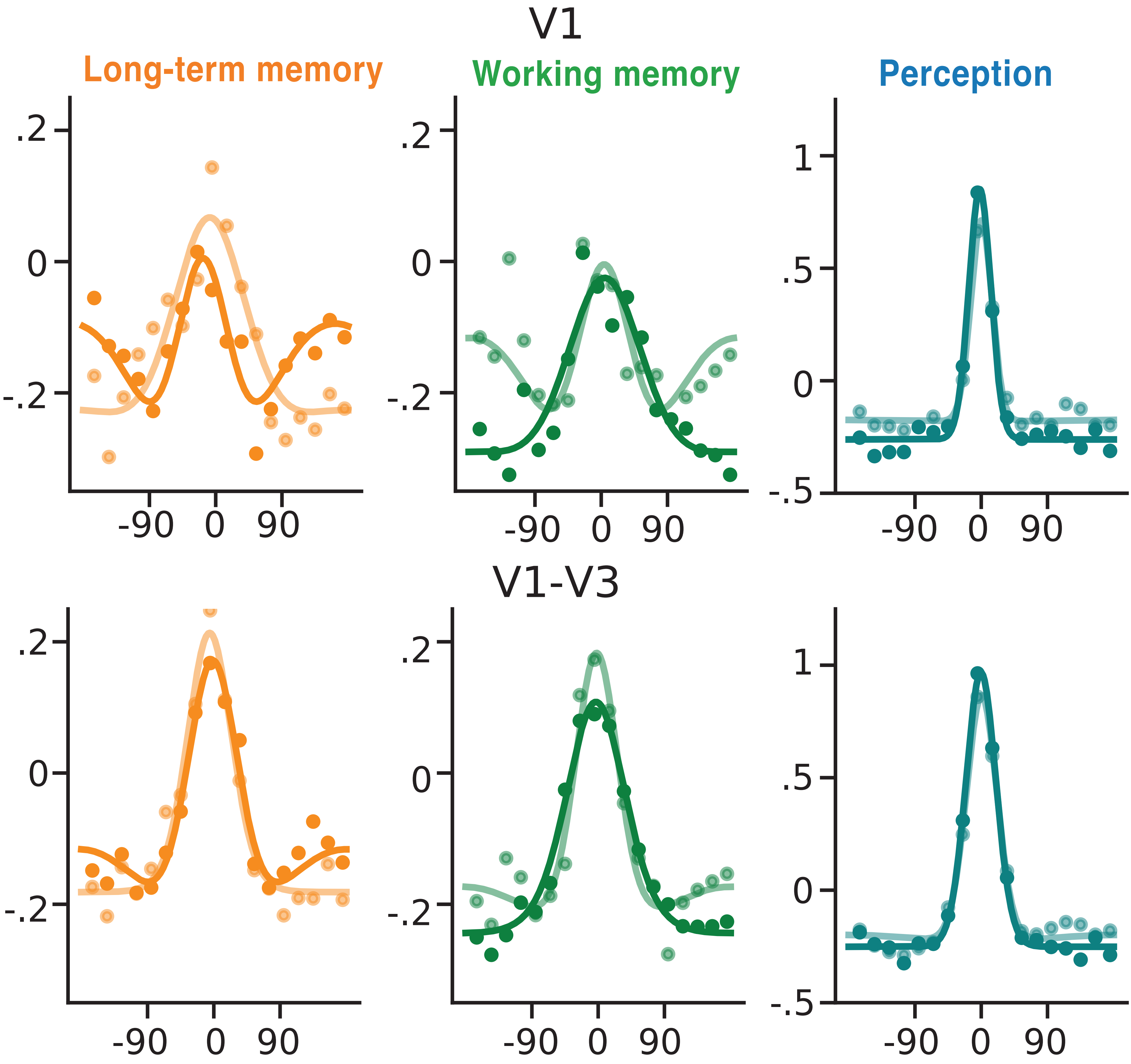
Cortical responses by hippocampal activity

Hippocampal activity is more strongly linked to the strength of population responses in long-term memory than in working memory



Cortical responses by prefrontal activity

Prefrontal activity is differentially linked to population responses in visual cortex during working memory and long-term memory



Hippocampus and prefrontal cortex differentially shape memory responses in early visual cortex

Hippocampus more strongly linked to stronger long-term memory response
Prefrontal cortex differently linked to working memory and long-term memory responses

There are multiple top-down pathways to shape sensory brain activity.

References

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